multiSpeciesRegionFoam

Physics Reference Guide

Species transport through solid membranes and porous materials
in the OpenFOAM-13 `foamMultiRun` framework

Scope. This document describes every physical model implemented in the `multiSpeciesRegionFoam` library: governing equations, material constitutive laws, interface and surface boundary conditions, performance metrics, and numerical discretisation. Each section indicates which tutorial cases (case311–case322) demonstrate the corresponding physics.

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Contents

1 Introduction and Scope **2**

2 Governing Equations **2**
2.1 Species conservation . . . 2
2.2 Energy conservation . . . 2
2.3 Species and energy coupling . . . 3

3 Material Models **3**
3.1 Arrhenius temperature dependence . . . 3
3.2 Solubility laws . . . 4
3.2.1 Sieverts' law (metals and interstitial solids) . . . 4
3.2.2 Henry's law (polymers, liquids, porous media) . . . 4
3.3 McNabb–Foster trapping kinetics . . . 4
3.3.1 Trapping regimes . . . 5
3.3.2 Numerical discretisation of the trapping ODE . . . 5

4 Interface and Surface Boundary Conditions **5**
4.1 Overview . . . 5
4.2 Sieverts–Sieverts interface (linear partition) . . . 5
4.3 Henry–Sieverts interface (nonlinear partition) . . . 6
4.3.1 Steady-state interface concentrations . . . 7
4.4 Surface recombination and dissociation . . . 7
4.4.1 Equilibrium and Sieverts limit . . . 7
4.4.2 Damköhler number . . . 8
4.4.3 Steady-state surface concentration . . . 8

5 Performance Metrics **8**
5.1 Permeation flux and speciesFlux function object . . . 8
5.2 Permeation reduction factor . . . 9

6 Numerical Implementation **10**
6.1 Finite-volume discretisation . . . 10
6.2 PIMPLE outer loop and Picard convergence . . . 10
6.3 Coupled-patch data exchange . . . 10
6.4 Semi-implicit trapping ODE . . . 10
6.5 Library architecture . . . 11

7 Tutorial Guide **11**
7.1 Case summary table . . . 11
7.2 Physics-to-tutorial cross-reference . . . 11
7.3 Running the tutorials . . . 11

8 Notation Summary **13**

References **13**

Introduction and Scope

`multiSpeciesRegionFoam` is an OpenFOAM-13 add-on that extends the native `foamMultiRun` [7] multi-region framework with coupled species transport. Each solid region runs the `speciesSolid` solver module, which adds a species conservation equation alongside the inherited thermal energy equation. Inter-region coupling (heat and species) uses OpenFOAM's mapped-patch infrastructure, and all new boundary conditions are registered in the standard run-time selection tables [1].

Physical scope. The library targets *diffusion-dominated* transport of a dilute dissolved species (hydrogen isotopes H/D/T, vapour, dissolved salt) through one or more solid layers. Convection inside the solid and binary diffusion effects are neglected; the driving force is the concentration gradient in each layer. Typical applications are:

- Hydrogen permeation through hydrogen carrier pipes and storage tanks
- Tritium permeation through first-wall and breeding blanket materials in fusion machines, and through structural materials in molten salt reactors (H, D, T transport; Sieverts-law solubility).
- Dense-film membrane separation: reverse osmosis, membrane distillation, pervaporation (Henry-law solubility, Arrhenius temperature dependence).
- Permeation barriers (tungsten carbide, alumina coatings) [4].
- Hydrogen permeation in heat exchangers with molten-salt coolants.

Governing Equations

Species conservation

In each solid region the *mobile* species concentration $C\,[\mathrm{mol\ m^{-3}}]$ satisfies:

Species equation — every solid region

$$\frac{\partial C}{\partial t} = \nabla \cdot (D(T)\,\nabla C) - \frac{\partial C_t}{\partial t} + S_{\mathrm{vol}} \tag{1}$$

Symbol	Meaning	Units
C	Mobile species concentration	$\mathrm{mol\ m^{-3}}$
$D(T)$	Temperature-dependent diffusivity	$\mathrm{m^2\ s^{-1}}$
C_t	Trapped concentration (§3.3)	$\mathrm{mol\ m^{-3}}$
S_{vol}	Uniform volumetric source	$\mathrm{mol\ m^{-3}\ s^{-1}}$

When trapping is disabled (`trappingModel noTrapping`), $\partial C_t/\partial t = 0$ and the equation reduces to pure diffusion. The source term S_{vol} is optional and uniform within each region.

Demonstrated by: all twelve tutorial cases. Cases `case311`–`case312` use the thermal field T as a species proxy to verify the pure-diffusion solver before species coupling is introduced.

Energy conservation

The thermal energy equation is inherited from OpenFOAM-13's `solid` module:

Energy equation (Fourier conduction)

$$\frac{\partial(\rho\, e)}{\partial t} + \nabla\cdot\mathbf{q} = S_e, \qquad \mathbf{q} = -\kappa(T)\,\nabla T \tag{2}$$

Symbol	Meaning	Units
ρ	Density	kg m^{-3}
e	Specific internal energy	J kg^{-1}
$\mathbf{q}$	Conductive heat flux	W m^{-2}
$\kappa(T)$	Thermal conductivity	W m^{-1} K^{-1}
S_e	Volumetric heat source	W m^{-3}

The species and energy equations are solved *sequentially* within each PIMPLE outer corrector. After the energy equation updates T, the diffusivity $D(T)$ (and all Arrhenius properties) are refreshed before assembling the species equation.

Full energy–species coupling: `case316` (membrane distillation with Arrhenius $D(T)$ driven by a temperature gradient), `case320` (three-region shell-and-tube heat exchanger; conjugate heat transfer through FLiBe–SS316–Hitec).
Isothermal (energy solved but T fixed): `case311`–`case315`, `case317`–`case319`, `case321`–`case322`.

Species and energy coupling

In multi-region cases, the PIMPLE loop exchanges temperature across coupled patches via `coupledTemperature` boundary conditions (standard OpenFOAM), and exchanges species concentration via the custom `sievertsCoupledMixed` or `surfaceRecombination` BCs described in §4. Both exchanges happen every PIMPLE outer corrector, so the fields remain thermodynamically consistent throughout the time step.

Material Models

Arrhenius temperature dependence

Every material property that depends on temperature follows the Arrhenius form. This applies uniformly to D, the solubility K_s, and the surface-kinetics coefficients K_d, K_r:

Arrhenius property

$$X(T) = X_0 \exp\left(-\frac{E_a}{RT}\right) \tag{3}$$

Symbol	Meaning	Notes
X_0	Pre-exponential factor	Carries the SI units of X
E_a	Activation energy	J mol^{-1}
R	Gas constant	8.314 J mol^{-1} K^{-1}
T	Local temperature	K

Setting $E_a = 0$ gives a temperature-independent constant, used in all isothermal tutorial cases.

Dictionary syntax (applies to all Arrhenius properties):

```
D { X0  [0 2 -1 0 0 0 0]  2.9e-7;   Ea  13800; }
```

Isothermal ($E_a = 0$): `case311`–`case315`, `case317`–`case319`, `case321`–`case322`.
Arrhenius $D(T)$ active: `case316` (D(T) for vapour diffusion), `case319` (real $D(T)$ for WC and SS316), `case320` (real $D(T)$ for SS316 wall at T gradient from FLiBe/Hitec).

Solubility laws

Sieverts' law (metals and interstitial solids)

Sieverts' law describes the equilibrium solubility of diatomic gases (H_2, D_2, T_2) that dissociate upon dissolution. The dissolved concentration is proportional to the *square root* of the gas partial pressure:

Sieverts' law

$$C \;=\; K_s(T)\,\sqrt{p} \tag{4}$$

$K_s(T)\,[\,\mathrm{mol\ m^{-3}\ Pa^{-1/2}}]$ is the Sieverts solubility constant, following the Arrhenius form (3). Applicable materials include iron, stainless steel (SS316), tungsten, vanadium alloys, and palladium [6].

Used in: `case312`–`case314`, `case318`–`case320`, `case321` (metal layer), `case322`.

Henry's law (polymers, liquids, porous media)

For species that dissolve without dissociation (vapour in a polymer, dissolved gas in a liquid), the equilibrium concentration is proportional to pressure:

Henry's law

$$C \;=\; K_H(T)\,p \tag{5}$$

$K_H(T)\,[\,\mathrm{mol\ m^{-3}\ Pa^{-1}}]$ is the Henry constant. Applicable to dense polymer films, membrane distillation membranes, and organic solvents [5].

Used in: `case315`–`case316` (membrane distillation), `case317` (reverse osmosis, $K_H = 0.5$ at the feed/membrane interface), `case321` (polymer layer).

McNabb–Foster trapping kinetics

In metals, hydrogen isotopes can occupy lattice defects (vacancies, dislocations, grain boundaries). The McNabb–Foster model [2] treats a single population of trap sites with density $n_t = f_t N\,[\,\mathrm{mol\ m^{-3}}]$:

McNabb–Foster ODE

$$\frac{\partial C_t}{\partial t} \;=\; \frac{\alpha_t}{N}\,C\,(n_t - C_t) \;-\; \alpha_d(T)\,C_t \tag{6}$$

Symbol	Meaning	Units
N	Molar trap-site density	mol m^{-3}
f_t	Trap fraction ($0 < f_t \leq 1$)	—
$n_t = f_t N$	Available trap-site density	mol m^{-3}
α_t	Trapping rate coefficient	s^{-1}
$\alpha_d(T) = \alpha_{d0} \exp(-\varepsilon/k_B T)$	Detrapping rate	s^{-1}
ε	Trap binding energy	eV
k_B	Boltzmann constant	J K^{-1}

The trapping term $\partial C_t/\partial t$ is subtracted from the mobile species equation (1) as a sink.

Trapping regimes

Two limiting regimes exist depending on the ratio of trapping to diffusion time scales:

- **Weak trapping** ($\alpha_t/D \ll 1$): traps fill slowly; the mobile front propagates at nearly the bare diffusion speed; breakthrough time $\approx L^2/D$.
- **Strong trapping** ($\alpha_t/D \gg 1$): most atoms are trapped; effective diffusivity $D_{\text{eff}} = D/(1 + N_t\,\alpha_t)$ is greatly reduced; breakthrough is delayed by orders of magnitude.

Numerical discretisation of the trapping ODE

The ODE (6) is stiff and must be discretised carefully to remain bounded ($0 \leq C_t \leq n_t$). A per-cell semi-implicit Euler step is used:

Semi-implicit Euler for trapping ODE

$$C_t^{n+1} = \frac{C_t^n + \Delta t\,\dfrac{\alpha_t}{N}\,C\,n_t}{1 + \Delta t\left(\dfrac{\alpha_t}{N}\,C + \alpha_d\right)} \tag{7}$$

This scheme is unconditionally stable and guarantees $0 \leq C_t^{n+1} \leq n_t$ for any time step.

Demonstrated by: `case313`. Two sub-cases span the weak and strong trapping regimes. The analytical breakthrough time is used to validate both limits.

Interface and Surface Boundary Conditions

Overview

At every coupled patch, two physical conditions must hold simultaneously:

1. **Flux continuity:** $D_A\,(\partial C_A/\partial n)_{\text{face}} = D_B\,(\partial C_B/\partial n)_{\text{face}}$
2. **Thermodynamic equilibrium:** determined by the solubility law on each side (§3.2).

These are implemented as a *mixed* (Robin) boundary condition in OpenFOAM, derived from the conjugate heat-transfer analogy.

Sieverts–Sieverts interface (linear partition)

When both sides follow Sieverts' law, the equilibrium condition at the interface face is:

Sieverts–Sieverts partition (`partition linear`)

$$\frac{C_A}{K_{s,A}} = \frac{C_B}{K_{s,B}} \qquad \Longleftrightarrow \qquad C_A = \frac{K_{s,A}}{K_{s,B}} C_B \tag{8}$$

The concentration ratio $K_{s,A}/K_{s,B}$ can be greater or less than unity, producing a concentration jump at the interface even in steady state. When $K_{s,A} = K_{s,B}$, the concentration is continuous.

Mixed-BC coefficients (self = region A, neighbour = region B):

$$\begin{aligned} \text{refValue} &= \frac{K_{s,A}}{K_{s,B}} C_{B,\text{cell}} \\ w &= \frac{D_B\,\delta_B^{-1}}{D_B\,\delta_B^{-1} + D_A\,\delta_A^{-1}(K_{s,A}/K_{s,B})} \\ C_{A,\text{face}} &= w \cdot \text{refValue} + (1-w)\,C_{A,\text{cell}} \end{aligned} \tag{9}$$

where δ^{-1} is the distance from the cell centre to the interface face.

Demonstrated by: `case314` (two-layer composite membrane, K_s ratio = 4; interface concentration jump of factor 4 at steady state), `case318` (code-to-code benchmark, equal K_s, continuous profile), `case319` (WC/SS316 barrier, large K_s ratio responsible for PRF), `case320` (SS316 wall; also `case321` reference sub-case).

Henry–Sieverts interface (nonlinear partition)

When a Henry-law material (polymer, liquid) interfaces with a Sieverts-law material (metal), the same partial pressure p_{int} must hold on both sides:

$$p_{\text{int}} = \frac{C_{\text{Henry}}}{K_H} = \left(\frac{C_{\text{Sieverts}}}{K_s}\right)^2$$

This gives two equivalent partition conditions depending on which side is considered "self":

Henry–Sieverts partition

$$\text{Sieverts side (`partition sqrt`):} \quad C_{\text{Sieverts}} = K_s\sqrt{\frac{C_{\text{Henry}}}{K_H}} \tag{10}$$

$$\text{Henry side (`partition quadratic`):} \quad C_{\text{Henry}} = K_H\left(\frac{C_{\text{Sieverts}}}{K_s}\right)^2 \tag{11}$$

Both sides must be set consistently in the `0/` boundary field files; they encode the same physics from opposite perspectives.

Linearised mixed-BC weight for the nonlinear cases. The weight (9) is derived by linearising the partition function at the current Picard iterate. For the `sqrt` mode (self = Sieverts, neighbour = Henry):

$$\left.\frac{\mathrm{d}C_s}{\mathrm{d}C_n}\right|_{\text{iterate}} = \frac{C_{s,\text{face}}}{2\,C_{n,\text{cell}}} \qquad \Rightarrow \qquad \text{selfKD} = D_A\,\delta_A^{-1}\,\frac{C_{A,\text{face}}}{2\,C_{B,\text{cell}}} \tag{12}$$

Using the linearised derivative (rather than the constant K_s/K_H) in the weight formula ensures that flux continuity is enforced exactly at every Picard step, even in steady state on a coarse mesh.

Steady-state interface concentrations

For the case of equal layer thicknesses L, equal diffusivities D, inlet concentration $C_0 = 1$ mol m^{-3}, and $K_s = K_H = 1$:

Analytical interface concentrations (equal D, L, K)

$$u^2 + u = 1, \quad u = C_{\text{Sieverts,int}}, \qquad \Rightarrow \qquad u = \tfrac{\sqrt{5}-1}{2} \approx 0.618 \text{ mol m}^{-3} \tag{13}$$

The *golden-ratio conjugate* $(\sqrt{5}-1)/2$ appears naturally because the steady-state flux equality combined with the $\sqrt{\cdot}$ partition gives exactly the quadratic $u^2 + u - 1 = 0$. The Henry (polymer) side holds $C_{\text{Henry,int}} = u^2 \approx 0.382$ mol m^{-3}, demonstrating that the *concentration is discontinuous and higher on the Sieverts (metal) side* — the defining signature of this partition law.

The steady-state flux in the Henry–Sieverts case is 23.6 % *higher* than the equivalent Sieverts–Sieverts case with the same D, L, and K:

$$J_{\text{HS}} = D\,\frac{C_{\text{Sieverts,int}}}{L} = \frac{(\sqrt{5}-1)/2}{L/D}\,C_0 \;\approx\; 1.236\,J_{\text{SS}} \tag{14}$$

Demonstrated by: `case321`. Two sub-cases (`sieverts_sieverts` reference and `henry_sieverts`) run with identical D, L, boundary values; the 23.6 % flux difference is confirmed analytically and numerically.

Surface recombination and dissociation

At a free metal surface exposed to a molecular gas, two kinetic processes compete:

Dissociative absorption: $H_2 \to 2H$, flux $= K_d(T)\,p_{\text{gas}}$
Recombinative desorption: $2H \to H_2$, flux $= K_r(T)\,C^2$

The net flux into the solid is the *surface recombination Robin BC*:

Surface recombination BC (`surfaceRecombination`)

$$D\,\frac{\partial C}{\partial n} \;=\; K_d(T)\,p_{\text{gas}} \;-\; K_r(T)\,C^2 \tag{15}$$

Symbol	Meaning	Units
$K_d(T)$	Dissociation rate (Arrhenius)	mol m^{-2} s^{-1} Pa^{-1}
$K_r(T)$	Recombination rate (Arrhenius)	m^4 mol^{-1} s^{-1}
p_{gas}	Gas-phase partial pressure	Pa

Both K_d and K_r follow the Arrhenius form (3). Setting $p_{\text{gas}} = 0$ models a vacuum or purge side (pure desorption).

Equilibrium and Sieverts limit

At equilibrium (zero net flux), $C = C_{\text{eq}}$:

$$K_d\,p_{\text{gas}} = K_r\,C_{\text{eq}}^2 \qquad \Rightarrow \qquad C_{\text{eq}} = \sqrt{\frac{K_d\,p_{\text{gas}}}{K_r}} \tag{16}$$

This is exactly Sieverts' law (4) with $K_s = \sqrt{K_d/K_r}$: *the surface recombination BC generalises the Sieverts equilibrium BC and recovers it in the fast-kinetics limit* [3].

Damköhler number

The Damköhler number compares the surface-kinetics rate to the bulk-diffusion rate:

Damköhler number

$$\mathrm{Da} = \frac{K_r\, C_{\mathrm{eq}}\, L}{D} \tag{17}$$

Da	Limiting step	Surface concentration
$\mathrm{Da} \gg 1$	Diffusion-limited	$C_s \approx C_{\mathrm{eq}}$ (Sieverts equilibrium)
$\mathrm{Da} \approx 1$	Mixed control	$C_s = \frac{\sqrt{5}-1}{2}\, C_{\mathrm{eq}}$ (see below)
$\mathrm{Da} \ll 1$	Kinetically limited	$C_s \ll C_{\mathrm{eq}}$

Steady-state surface concentration

At steady state, the Robin condition (15) must equal the permeation flux $D\, C_s/L$ (assuming zero concentration at the purge face):

$$K_r\, C_s^2 + \frac{D}{L}\, C_s - K_d\, p_{\mathrm{gas}} = 0 \tag{18}$$

The physical (positive) root is:

Steady-state surface concentration (general Da)

$$C_s = \frac{-D/L + \sqrt{(D/L)^2 + 4\, K_r\, K_d\, p_{\mathrm{gas}}}}{2\, K_r} \tag{19}$$

Special case $\mathrm{Da} = 1$ (with $K_d = K_r$, $p_{\mathrm{gas}} = 1$, $D/L = K_r\, C_{\mathrm{eq}} = K_r$):

$$C_s^2 + C_s - 1 = 0 \qquad \Rightarrow \qquad C_s = \tfrac{\sqrt{5}-1}{2} \approx 0.618 \text{ mol m}^{-3} \tag{20}$$

The golden-ratio conjugate appears again. The flux is reduced by 38.2 % relative to the Sieverts equilibrium limit:

$$\frac{J_{\mathrm{surf}}}{J_{\mathrm{ref}}} = \frac{C_s}{C_{\mathrm{eq}}} = \frac{\sqrt{5}-1}{2} \approx 0.618 \tag{21}$$

Numerical treatment. The C^2 term is Picard-frozen at the previous cell concentration, making the gradient a simple field evaluation each PIMPLE outer iteration. Convergence is guaranteed by the PIMPLE outer loop.

Demonstrated by: `case322`. Sub-case `sieverts_bc` provides the $\mathrm{Da} \to \infty$ reference (J_{ref}); sub-case `surface_recombination` runs $\mathrm{Da} = 1$ and confirms $C_s = (\sqrt{5}-1)/2$ and the 38.2 % flux reduction.

Performance Metrics

Permeation flux and speciesFlux function object

The area-integrated molar permeation flux through a patch is:

Surface molar flux

$$J_{\text{patch}} = -\int_{\text{patch}} D\,\nabla C \cdot \hat{n}\,\mathrm{d}A \qquad [\,\text{mol s}^{-1}] \tag{22}$$

The `speciesFlux` function object evaluates this integral at every write time without modifying the solver. It outputs both the area-averaged flux density $[\,\text{mol m}^{-2}\ \text{s}^{-1}]$ and the total flux $[\,\text{mol s}^{-1}]$ to `postProcessing/<name>/<time>/speciesFlux.dat`.

The diffusivity field `D_<species>` is registered as `AUTO_WRITE` by the `speciesModel`, making it available to the function object at every write time.

Dictionary syntax:

```
functions
{
    wallPermeation
    {
        type        speciesFlux;
        libs        ("libspeciesPost.so");
        region      wall;
        species     C_H2;
        patches     (wall_to_hitec);
        writeControl writeTime;
    }
}
```

Demonstrated by: `case320` (permeation flux through SS316 wall into Hitec coolant), `case321`, `case322` (permeation flux at purge face).

Permeation reduction factor

The permeation reduction factor (PRF) quantifies the effectiveness of a barrier coating:

Permeation reduction factor

$$\text{PRF} = \frac{J_{\text{without barrier}}}{J_{\text{with barrier}}} = \frac{C_0/R_{\text{total,bare}}}{C_0/R_{\text{total,coated}}} \tag{23}$$

For a two-layer composite (coating thickness L_c, substrate thickness L_s) with equal K_s on both sides, the steady-state PRF can be approximated by the ratio of total diffusion resistances. High PRF arises when the coating has a much larger K_s/D ratio (lower permeability) than the substrate.

The permeability of a material is $\Phi = K_s^2 D$ (in $[\,\text{mol m}^{-1}\ \text{s}^{-1}\ \text{Pa}^{-1}]$, the Richardson–Dushman formula for diatomic gases).

Demonstrated by: `case319`. A tungsten carbide (WC) coating on SS316 at $T = 1073$ K:

$$\text{PRF} \approx 200$$

confirmed by comparing the `with_coating` and `no_coating` sub-cases. The PRF is dominated by the large K_s ratio ($K_{s,\text{WC}} \gg K_{s,\text{SS316}}$), which lowers the dissolved concentration in the coating and therefore reduces the flux into the substrate.

Numerical Implementation

Finite-volume discretisation

The species equation (1) is discretised using the standard OpenFOAM finite-volume method on an unstructured collocated mesh:

- **Time integration:** first-order implicit Euler (`ddtSchemes: Euler`).
- **Diffusion:** second-order Gauss linear with explicit non-orthogonal correction (`laplacianSchemes: Gauss linear corrected`).
- **Gradient:** Gauss linear (`gradSchemes: Gauss linear`).

These settings are appropriate for diffusion-dominated transport on orthogonal structured meshes (all tutorial cases use 1-D block meshes).

PIMPLE outer loop and Picard convergence

The nonlinear boundary conditions (`quadratic`, `sqrt`, `surfaceRecombination`) are linearised by Picard (fixed-point) iteration within the PIMPLE outer corrector loop:

1. **Outer iteration k:** the nonlinear BC evaluates the interface condition using cell values from iteration $k-1$ (frozen coefficients).
2. **Species equation:** assembled and solved with the frozen coefficients; cell values are updated.
3. **Convergence check:** if the species residual $< \varepsilon$ (`residualControl`), the outer loop exits; otherwise another outer iteration is performed.

The mixed-BC weight (12) uses the *linearised derivative* of the partition function evaluated at the current iterate, ensuring that flux continuity is enforced at every Picard step and that the method converges to the exact partition condition as $k \to \infty$.

Coupled-patch data exchange

At each outer PIMPLE iteration, the coupled patches exchange:

- **Neighbour cell concentration** $C_{n,\text{cell}}$ (used to evaluate the partition refValue),
- **Neighbour diffusivity** D_n and **inverse distance** δ_n^{-1} (used to evaluate nbrKD for the weight).

Data exchange uses OpenFOAM's mapped-patch infrastructure with an incremented MPI message tag to avoid collisions when multiple coupled fields are exchanged in the same iteration.

Semi-implicit trapping ODE

The McNabb–Foster ODE (6) is solved *cell by cell* with the per-cell semi-implicit Euler update (7). This is performed explicitly before the species equation assembly, so the sink term $\partial C_t/\partial t$ is available for the right-hand side. No global matrix inversion is required for the ODE update.

Library architecture

Library	Purpose	Key classes
`libspeciesTransport.so`	Core physics	`speciesModel`, `arrheniusProperty`, `McNabbFoster`, `noTrapping`
`libspeciesCoupling.so`	Boundary conditions	`sievertsCoupledMixed`, `surfaceRecombination`
`libspeciesSolid.so`	Solver module	`speciesSolid`
`libspeciesPost.so`	Post-processing	`speciesFlux`

All libraries must be listed in `system/controlDict`:

```
libs ("libspeciesTransport.so" "libspeciesCoupling.so"
      "libspeciesSolid.so"    "libspeciesPost.so");
```

Tutorial Guide

The twelve tutorial cases are ordered by increasing physical complexity. Each case contains an `Allrun` script, an `Allclean` script, a per-case `README.md`, and a Python validation script that compares the simulation output to an analytical or benchmark solution.

Case summary table

Physics-to-tutorial cross-reference

Running the tutorials

Prerequisites. Source OpenFOAM-13 and build the libraries:

```
source /opt/openfoam13/etc/bashrc
cd $WM_PROJECT_USER_DIR/multiSpeciesRegionFoam
./Allwmake 2>&1 | tee build.log
```

Individual case:

```
cd tutorials/case314-composite-membrane
./Allrun
python3 validation/membrane_compare.py --plot
```

Validation exit code: each validation script exits 0 on PASS (flux within 5 % of analytical) and 1 on FAIL.

- **Fusion energy** — preventing tritium permeation through first-wall and blanket structures into the coolant

Notation Summary

Table 1: Tutorial cases and the physics they demonstrate.

Case	Solver	Physics / Key feature
`case311`	`solid`	Pure 1-D diffusion, step BC. Uses temperature field as species proxy. Validates erfc front propagation (exact analytical solution).
`case312`	`speciesSolid`	1-D diffusion, non-uniform initial condition (`setFields`). Fourier cosine series analytical solution. First use of the species equation.
`case313`	`speciesSolid`	McNabb–Foster trapping (§3.3). Two sub-cases: weak trapping ($\alpha_t/D \ll 1$) and strong trapping ($\alpha_t/D \gg 1$). Validates semi-implicit ODE (7).
`case314`	`speciesSolid`	Two-region composite membrane; Sieverts–Sieverts linear partition (§4.2) with $K_{s,A}/K_{s,B} = 4$. Validates concentration jump at the interface.
`case315`	`speciesSolid`	Isothermal membrane distillation. Henry-law solubility (§3.2), constant D. Linear steady-state profile.
`case316`	`speciesSolid`	Membrane distillation with Arrhenius $D(T)$ (§3.1). Temperature gradient drives non-linear steady-state profile. Full energy–species coupling active.
`case317`	`speciesSolid`	Reverse osmosis: two-region feed/membrane. Henry partition at the feed face ($K_H = 0.5$). Solution-diffusion model.
`case318`	`speciesSolid`	Code-to-code benchmark against Pasler et al. and Fourier sine series. Constant BCs; transient diffusion with exact analytical solution.
`case319`	`speciesSolid`	WC coating + SS316 substrate; real Arrhenius $D(T)$ for both materials. Linear partition ((8)). PRF ≈ 200 (§5.2). Two sub-cases: `with_coating` / `no_coating`.
`case320`	`speciesSolid`	Three-region shell-and-tube heat exchanger: FLiBe (shell) | SS316 (wall) | Hitec (tube). Full conjugate heat transfer + species permeation through the wall. Arrhenius $D(T)$ from T gradient through wall. `speciesFlux` function object measures J through the wall.
`case321`	`speciesSolid`	Henry–Sieverts partition (§4.3, §4.2). Two sub-cases: `sieverts_sieverts` (reference) and `henry_sieverts` (sqrt/quadratic BC). Validates 23.6 % flux increase and interface concentration jump. Golden-ratio interface concentration (13).
`case322`	`speciesSolid`	Surface recombination/dissociation (§4.4). Two sub-cases: `sieverts_bc` (Da $\to \infty$ reference) and `surface_recombination` (Da = 1). Validates golden-ratio surface concentration (20) and 38.2 % flux reduction.

Table 2: Which tutorials exercise which physical models.

Physical model	*311*	*312*	*313*	*314*	*315*	*316*	*317*	*318*	*319*	*320*	*321*	*322*
Species diffusion eq. (1)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Energy eq. (2)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Arrhenius (3)						✓			✓	✓		
Sieverts law (4)			✓	✓				✓	✓	✓	✓	✓
Henry law (5)					✓	✓	✓				✓	
McNabb–Foster (6)			✓									
Linear partition (8)				✓				✓	✓	✓	✓	
Sqrt partition (10)											✓	
Quadratic partition (11)							✓				✓	
Surface recombination (15)												✓
PRF metric (23)									✓			
speciesFlux FO (22)										✓	✓	✓
Multi-region (≥ 3 regions)										✓		

Symbol	Meaning	SI units
C	Mobile species concentration	mol m^{-3}
C_t	Trapped concentration	mol m^{-3}
$D(T)$	Diffusivity (Arrhenius)	m^2 s^{-1}
E_a	Activation energy	J mol^{-1}
f_t	Trap fraction	—
J	Molar flux (area-averaged)	mol m^{-2} s^{-1}
$K_d(T)$	Dissociation rate	mol m^{-2} s^{-1} Pa^{-1}
$K_H(T)$	Henry solubility constant	mol m^{-3} Pa^{-1}
$K_r(T)$	Recombination rate	m^4 mol^{-1} s^{-1}
$K_s(T)$	Sieverts solubility constant	mol m^{-3} Pa$^{-1/2}$
L	Layer thickness	m
N	Molar trap-site density	mol m^{-3}
$n_t = f_t N$	Available trap density	mol m^{-3}
p	Gas-phase partial pressure	Pa
R	Gas constant (8.314)	J mol^{-1} K^{-1}
S_{vol}	Volumetric species source	mol m^{-3} s^{-1}
T	Temperature	K
w	Mixed-BC value fraction	—
X_0	Arrhenius pre-exponential factor	(same as X)
$\alpha_d(T)$	Detrapping rate	s^{-1}
α_t	Trapping rate	s^{-1}
δ^{-1}	Cell-centre to face distance (inv.)	m^{-1}
ε	Trap binding energy	eV
$\kappa(T)$	Thermal conductivity	W m^{-1} K^{-1}
$\Phi = K_s^2 D$	Permeability (diatomic gas)	mol m^{-1} s^{-1} Pa^{-1}
Da	Damköhler number (surface kinetics)	—
PRF	Permeation reduction factor	—

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